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Assignment 1

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Program: Data Science and Informaion Technologies

Specialization: Bioinformatics - Biomedical Data

Lesson: Machine Learning In Computational Biology

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Contents

1 Abstract

In recent years the role of the gut microbiome in health ?? has been elucidated.

2 Introduction

In this assignment we try to compare different predictors for *BMI* depending on the prevalence and diversity of the gut microbiome.

3 Methods

3.1 Models

Three different regressors were used in the analysis.

Models:

1. Bayesian and Ridge
2. Support Vector Regression
3. Elastic Net

Models were imported from the *sklearn* library in *Python*.

3.2 Evaluation metrics

The "" were chosen as evaluation metrics. "" because they are listed as mandatory and "" because we have seen the distributions of the pairs of features and we know:

1. The categorical data is not going to be in the final data, but in the final metadata.
2. Some sets of features contain what seem like outliers.

As a result of knowing about outliers a median metric must be included to compare with the non-median metrics. Since all of our data will be of continuous variables we know the set of all possible metrics and the R^2 metric is the default.

3.3 Confidences

1000 iteration bootstrapping was done on models for evaluation along with k-fold cross validation. The *sklearn* library does not contain a bootstrapping function that we can use to call on our models if the model instance object does not already contain a method for bootstrapping, a protocol was written at [our source files](#).

All files can be found at [this repository](#).

4 Results

5 Disclaimer

Large language models (LLM's) were used during the assignment.

5.1 LLM

Open AI: Chat GPT 4.0

5.2 Purpose

1. To check ideas, like if the z-scaled data must be zero mean and unit variance (apparently it is) or if it's conditional (or to explain ideas).
2. To ask if concepts already exist in some dependency, as was the case for the way we cast the labels of the sex feature in the original data to binary representations (also re-used for the Project ID).

3. For acquisition of the relative documentation.

4. For helping with the L^AT_EXtable formatting in this report for time effi-

6 Citations